Knowledge Graph vs Retrieval-Augmented Generation for Legal Prior Case Retrieval

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***Abstract*—Legal prior case retrieval is a critical task in judicial systems where relevant precedents must be identified from large collections of legal documents. This work compares Knowledge Graph (KG), Retrieval-Augmented Generation (RAG), and hybrid approaches. Results show that semantic retrieval methods outperform structural methods after removing data leakage, while hybrid approaches provide balanced performance.**

***Index Terms*—Legal NLP, Knowledge Graph, RAG, LLM, Information Retrieval**

I. Introduction

Legal systems rely heavily on precedents, making prior case retrieval an essential component of legal research. Given a query case, the objective is to identify relevant past cases that influence judicial decisions. Traditional approaches rely on keyword matching or semantic similarity, which often fail due to the complex and domain-specific nature of legal language.

Recent advancements in Natural Language Processing (NLP) and Deep Learning have introduced methods such as Retrieval-Augmented Generation (RAG) and Knowledge Graph-based reasoning. While RAG leverages textual similarity and embeddings, Knowledge Graphs capture structured relationships such as shared statutes, judges, and citations.

This work explores and compares these approaches on the IL-TUR Prior Case Retrieval dataset, analyzing their effectiveness and limitations.

II. Literature Review

Earlier legal retrieval systems relied on rule-based and keyword matching approaches, which lacked semantic understanding. With the introduction of machine learning, TF-IDF and BM25-based retrieval improved performance but still suffered from vocabulary mismatch Deep learning approaches using embeddings and transformer-based models enabled better semantic representation of legal documents. The RA-KG-LLM framework further integrates knowledge graphs with LLM-based retrieval, combining structural and textual reasoning.

III. Objectives

The project goals are:

- To construct a knowledge graph from legal case documents.
- To implement RAG-based and KG-based retrieval systems.
- To compare multiple retrieval approaches using standard evaluation metrics.
- To analyze the impact of structural vs semantic features.
- To identify the most effective method for legal case retrieval.

IV. Proposed Methodology

A. Overview

The proposed system retrieves relevant prior legal cases by combining structural relationships and semantic similarity. Knowledge Graphs capture relationships such as shared statutes and judges, while Retrieval-Augmented Generation (RAG) captures the semantic meaning of legal text. By integrating these approaches along with neural re-ranking and fusion techniques, the system improves retrieval accuracy and provides more reliable results.

B. Data Preparation

The dataset is cleaned and normalized to remove noise and inconsistencies. The text is tokenized and processed for analysis. Important legal entities such as statutes, judges, courts, and case identifiers are extracted using rule-based methods. The processed text is then converted into numerical representations using TF-IDF and dense embeddings to capture both basic and semantic similarity.

Key Steps:

- Cleaning and normalization of text
- Tokenization and preprocessing

- Extraction of legal entities (statutes, judges, courts)
- Conversion to TF-IDF and dense embeddings

C. Knowledge Graph Construction

A knowledge graph is constructed where each case and extracted entity is represented as a node, and edges represent relationships such as shared statutes, judges, or courts. This helps capture structural connections between cases beyond simple text similarity.

Key Features:

- Nodes represent cases and legal entities
- Edges represent relationships (shared statutes, judges, courts)
- Captures structural dependencies between cases

D. Retrieval-Augmented Generation

Documents are converted into vector representations and indexed using FAISS for efficient similarity search. Given a query case, the system retrieves the most relevant cases based on textual similarity.

Approaches Used:

- TF-IDF (baseline similarity)
- Dense embeddings (semantic similarity)

E. RA-KG-LLM Approach

This approach combines knowledge graph information with text-based retrieval. Graph relationships are converted into triples and vectorized. The system retrieves both similar documents and graph patterns, improving contextual understanding.

Key Steps:

- Convert graph relationships into triples
- Vectorize triples
- Retrieve similar triples and documents
- Combine structural and semantic features

F. Neural Re-ranking

A cross-encoder model is used to re-rank retrieved candidates by comparing the query case and candidate cases together and assigning a relevance score. This improves ranking accuracy, especially for top results.

Key Idea:

- Input: query case + candidate case
- Output: relevance score

G. Hybrid Fusion

Results from different methods are combined using Reciprocal Rank Fusion (RRF) to produce the final ranked output.

Fusion Includes:

- Knowledge Graph results
- RAG results
- RA-KG-LLM results

H. Overall Workflow

1) Input query case
2) Preprocess text and extract entities
3) Generate representations (TF-IDF / embeddings)
4) Retrieve candidates using Knowledge Graph and RAG
5) Enhance retrieval using RA-KG-LLM
6) Re-rank results using neural cross-encoder model
7) Combine rankings using Reciprocal Rank Fusion (RRF)
8) Output top relevant prior cases

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6) Re-rank using neural model
7) Combine results using RRF
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V. Methodology Workflow

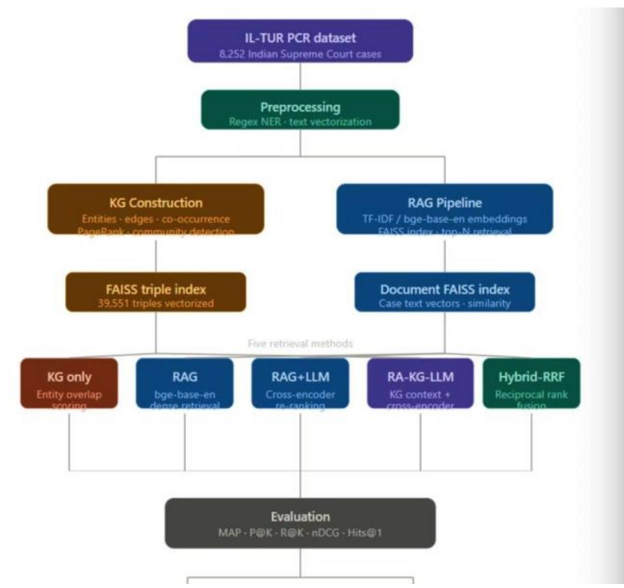

Fig. 1. Overall system workflow combining KG, RAG, and hybrid methods

VI. Results and Analysis

A. Experimental Setup

Experiments were conducted using Python, NumPy, Pandas, and FAISS. Data leakage was removed for fair evaluation.

B. Evaluation Metrics

- MAP
- Precision, Recall, F1-score
- nDCG
- Hits@K

C. Key Findings

As shown in Fig. 2, RAG-based methods outperform Knowledge Graph approaches. Hybrid methods provide stable performance.

Table I summarizes the performance at K=10. RAG-BM25 achieves the highest scores, while KG shows the lowest performance.

TABLE I
PERFORMANCE COMPARISON AT K=10

Method	F1@10	nDCG@10
KG	0.087	0.109
RAG-BM25	0.266	0.347
RAG+LLM	0.232	0.297
RA-KG-LLM	0.127	0.148
Hybrid-RRF	0.254	0.334

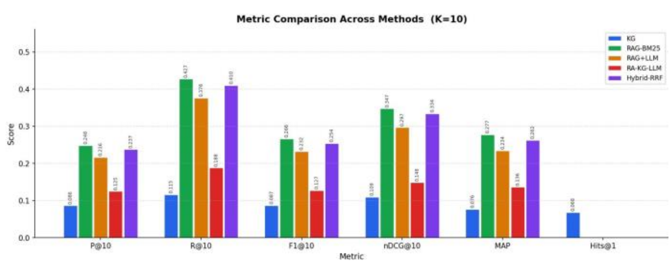

Fig. 2. Metric-wise comparison across methods

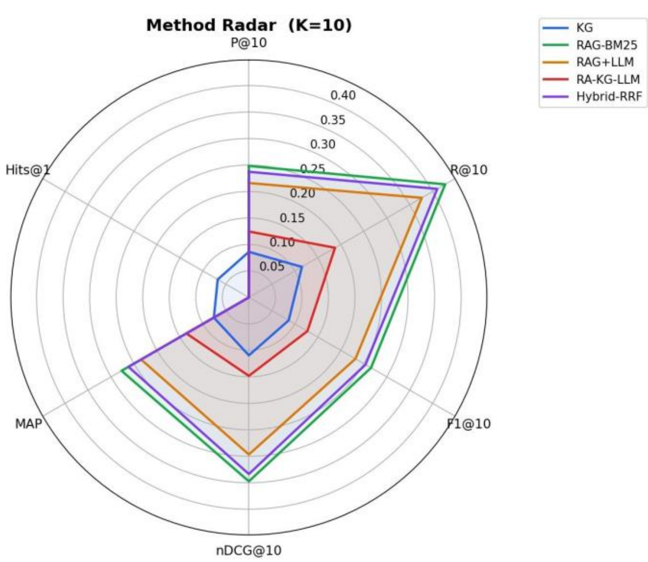

Fig. 3. Radar comparison of all methods

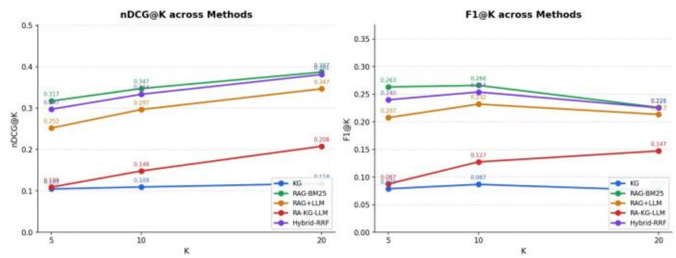

Fig. 4. nDCG comparison across methods

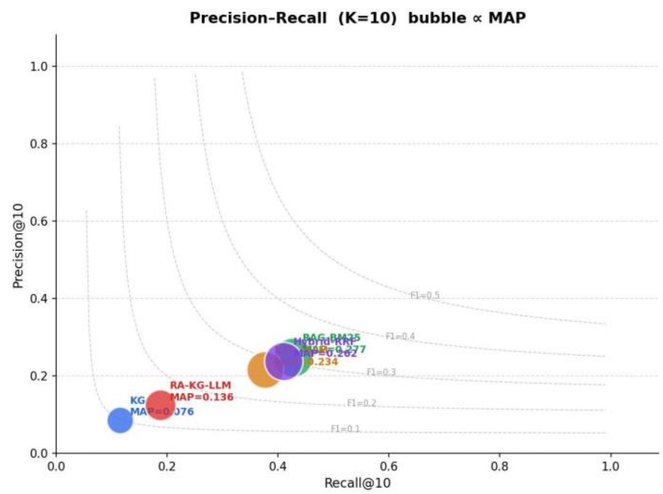

Fig. 5. Precision-Recall analysis across methods

D. Result Interpretation

- Knowledge Graph: Low performance due to lack of semantic understanding.
- RAG-BM25: Best performance.
- RA-KG-LLM: Moderate performance.
- Hybrid-RRF: Balanced performance.

VII. COMPARATIVE ANALYSIS

The comparative analysis shows clear differences between the methods. Knowledge Graph (KG) performs poorly as it relies only on structural relationships and lacks semantic understanding. RAG-based methods achieve the best performance by capturing textual similarity between legal cases.

The RA-KG-LLM approach improves over KG by combining structure and semantics but still does not match RAG performance. The Hybrid-RRF method provides balanced results by integrating all approaches.

Overall, semantic methods outperform structural methods, while hybrid approaches offer more stable and reliable performance.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

This work presented a comparative study of Knowledge Graph (KG), Retrieval-Augmented Generation (RAG), and hybrid approaches. The results show that KG-based methods are limited without data leakage, while RAG methods perform better by capturing semantic similarity. Hybrid approaches provide balanced and reliable performance.

B. Future Work

Future work includes using domain-specific models such as Legal-BERT, improving LLM-based re-ranking, and exploring graph-enhanced retrieval methods like GraphRAG. Deploying the system in real-time applications can further improve its usefulness.

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